

Class-1: Hydrozoa

They may be solitary (e.g. *Hydra*) or colonial. Life history includes sedentary, asexual, polypoid form and free swimming, sexual, medusoid form alternating with each other (e.g. *Obelia*). Mesoglea is non-cellular. Coelenteron is undivided (simple). Cnidocytes (cnidoblasts) occur only in the ectoderm. Germ cells are derived from the ectoderm. Some exhibit *polymorphism* with different types of **zooids** e.g. *Physalia*.

Examples: *Hydra* (freshwater polyp), *Obelia* (sea fur), *Physalia* (Portuguese man-of-war). Note: Zooid is a single individual in a colony.

Do you know why *Physalia* is commonly called ‘Portuguese man-of-war’ or ‘man o’ war’? The word ‘Man of War’ means a ‘battle ship with cannons and other powerful weapons’ Portugal once had. *Physalia* has many tentacles provided with venomous ‘stinging cells’, that are harmful to other organisms, hence the name ‘Portuguese man-of-war’.

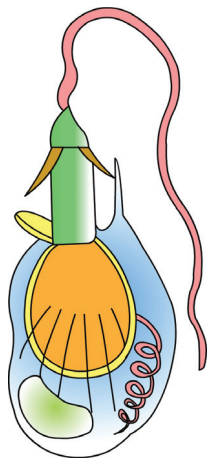


Fig. 3.3 Diagrammatic view of cnidoblast

Class-2: Scyphozoa

They are commonly referred to as jelly fishes as they possess a large quantity of jelly like mesoglea. They are solitary and medusoid in form. Mouth is surrounded by four oral arms. Coelenteron is divided into four chambers and some canals. Mesoglea contains amoebocytes. Cnidocytes occur in ectoderm and endoderm. Germ cells are derived from the endoderm.

Examples: *Aurelia* (jelly fish or moon jelly), *Rhizostoma*.

Class-3: Anthozoa or Actinozoa

They are commonly referred to as *sea anemones*. They are sedentary and *only* polypoid in form. Coelenteron is divided into several compartments by vertical septa called mesenteries. Mesoglea contains connective tissue. Cnidocytes occur both in the ectoderm and endoderm. Germ cells are derived from the endoderm.

Examples: *Adamsia* (sea anemone), *Corallium rubrum* (precious red stone coral), *Gorgonia* (sea fan), *Pennatula* (sea pen).

3.3 PHYLUM-CTENOPHORA

- Close relatives of cnidarians

3.3.1 General Characters

Ctenophores, commonly known as ‘**sea walnuts**’ or ‘**comb jellies**’ or ‘**sea gooseberries**’, are exclusively marine, **radially symmetrical**, diploblastic organisms with tissue level of organisation. The body bears *eight* external rows of ciliated **comb plates**, which help in locomotion, hence the name ‘ctenophora’. Cnidocytes are absent. However, they possess ‘**glue cells**’ called

